Experiment No: 6

Date:

Experiment Title: Design of ATM System using UML Use Case Diagram

Aim

To design and explain an Automated Teller Machine (ATM) system using a UML Use Case Diagram by identifying the main actors and showing how they interact with the system during ATM transactions.

Introduction

UML (Unified Modeling Language) is used to represent the working of a system in a simple and visual way. A Use Case Diagram shows what a system does and how users interact with it. It helps in understanding system functions clearly without going into technical details.

An ATM system allows customers to access banking services without visiting the bank. Common services include cash withdrawal, balance inquiry, and receipt printing. The Use Case Diagram helps in understanding these services and the role of each component involved in the ATM system.

Description of ATM Use Case Diagram

The ATM Use Case Diagram explains the interaction between users and different parts of the ATM system. Each use case represents a function performed by the system. The diagram focuses on user actions and system responses.

Actors Involved

1. Customer

The Customer is the main user of the ATM system. The customer inserts the ATM card, enters the PIN, selects the transaction type, withdraws cash, and collects the receipt.

2. ATM Interface

The ATM Interface is the part of the ATM that the customer interacts with. It includes the screen, keypad, card reader, and cash dispenser. It accepts input from the customer and shows messages on the screen.

3. Bank Server

The Bank Server processes all customer requests. It verifies the PIN, checks the account balance, approves transactions, and sends responses to the ATM Interface.

4. Bank Database

The Bank Database stores customer details such as account number, PIN, balance, and transaction records. It is accessed by the Bank Server during every transaction.

Use Cases

1. Insert ATM Card

The customer inserts the ATM card into the machine. The ATM Interface reads the card details and starts the transaction process.

2. Authenticate User

The customer enters the PIN. The ATM Interface sends the PIN to the Bank Server, which checks it with the Bank Database. If the PIN is correct, the user is allowed to continue.

3. Select Transaction

After successful authentication, the customer selects the required transaction such as cash withdrawal or balance inquiry.

4. Check Account Balance

Before withdrawing money, the Bank Server checks the account balance using the Bank Database and sends the information to the ATM Interface.

5. Withdraw Cash

The customer enters the withdrawal amount. If sufficient balance is available, the ATM Interface dispenses cash.

6. Update Account

After cash withdrawal, the Bank Server updates the new balance in the Bank Database.

7. Print Receipt

The ATM Interface prints a receipt showing transaction details like amount withdrawn and remaining balance.

8. Eject ATM Card

After completing the transaction, the ATM Interface returns the ATM card to the customer.

Relationship Between Use Cases

- User authentication is required before any transaction.
- Cash withdrawal depends on balance checking.

Advantages of Use Case Diagram in ATM System

- Clearly represents system functionality from the user's perspective.
- Helps identify all possible interactions between users and the ATM system.
- Simplifies requirement analysis and system understanding.
- Serves as a foundation for designing other UML diagrams like Class and Sequence Diagrams.
- Enhances communication between developers, designers, and stakeholders.

Result

The ATM system was successfully modeled using a UML Use Case Diagram. All actors and use cases were clearly identified and explained in a simple manner.

Conclusion

The Use Case Diagram clearly explains how different actors interact with the ATM system. By including Customer, ATM Interface, Bank Server, and Bank Database, the diagram gives a complete view of the ATM transaction process in a simple and understandable way.